

Kshitij Raj

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Education

Shobhit Institute of Engineering & Technology

Aug. 2023 – June 2027

Bachelor of Technology in Computer Science and Engineering

Meerut, Uttar Pradesh

Relevant Coursework: Data Structures, Algorithm Analysis, Software Methodology, Database Management Systems, Computer Networks, Internet Technology, Web Development, Computer Architecture, Cryptography

Experience

Student Intern | *Scientific Analysis Group (SAG), DRDO*

Jan 2026 – Present

- Investigating the implementation of lattice-based cryptographic algorithms on hardware with **less than 1MB of SRAM**, ensuring functional parity with standard implementations.
- Successfully adapted 4+ complex C/C++ libraries for Arduino-based microcontrollers, prioritizing firmware stability and cross-hardware compatibility.
- Orchestrated low-level data handling and communication protocols between microcontrollers and peripheral interfaces to ensure secure data transmission.

Freelance MERN Developer

April 2025 – July 2025

Client Project: "Nari Vastaram" – E-commerce Platform for Indian Women's Clothing

Remote

- Freelanced a full-stack e-commerce platform using React.js, Node.js, Express.js, and MongoDB for an Indian women's clothing brand, managing 50+ SKUs and handling 100% of launch traffic.
- Delivered a mobile-first React.js UI with dynamic product filtering, a secure payment gateway integration and responsive checkout flow — deployed on Hostinger VPS.
- Architected RESTful APIs with Express.js for product catalog, cart, and order management — handling concurrent requests across the full shopping flow.
- Reduced client's operational overhead by building a custom Node.js admin dashboard for real-time inventory updates, order tracking, and sales visibility — eliminating manual management.

Projects

School Management System (ERP) | *Typescript, Node.js, MySQL, Express.js, Three.js*

Sept 2025 – Jan 2026

- Built a comprehensive ERP solution featuring multi-level authentication for Admin, Teachers, and Students, streamlining academic and administrative workflows.
- Implemented dynamic modules for attendance tracking, fee management with automated invoicing, and a centralized examination grading system.
- Designed and managed a relational database schema supporting 1,000+ student records in **MySQL**.
- GitHub Link**

WasteEx | *Typescript, Node.js, MongoDB, Blockchain, Cloudinary*

April 2025 – July 2025

- Developed a full-stack B2B marketplace using Node.js, Express.js, MongoDB, and React.js, enabling businesses to list and purchase industrial waste as raw materials.
- Built a real-time chat dashboard with Socket.IO and integrated blockchain-based smart contracts with an escrow payment system for secure, trustless transactions.
- Implemented Cloudinary-based media uploads and RESTful APIs to manage listings, contracts, and user profiles across buyer and seller roles.
- GitHub Link**

Blog Loom | *React.js, Node.js, MongoDB, AWS*

Jan 2025 – March 2025

- Built a full-stack CMS using the MERN stack (MongoDB, Express.js, React.js, Node.js) with rich-text editing, AWS S3 image hosting, and category-based post filtering.
- Engineered a React.js frontend with dynamic comment system, SEO-friendly routing, and responsive layout — improving content discoverability across devices.
- Secured the platform with JWT authentication, bcrypt password hashing, and protected API routes in Express.js to prevent unauthorized access.
- GitHub Link**

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C, C++, PHP

Frontend: Reactjs, HTML5, CSS

Backend: Node.js, Express.js

Databases: MySQL, PostgreSQL, MongoDB

DevOps & Cloud: Docker, CI/CD, AWS Cloud(S3), VPS Hosting (Hostinger)

Tools & Platforms: Git, GitHub, Postman, Linux, VS Code, Jupyter, Google Colab

API/Communication: RESTful APIs, WebSockets (Socket.IO)

Other: Blockchain, Cloudinary

Soft Skills

- **Project Management:** Experienced in the full Software Development Life Cycle (SDLC), from client requirement gathering to post-deployment maintenance.
- **Technical Leadership:** Mentored peers in MERN stack development; winner of university-level hackathons focusing on privacy-centric solutions.
- **Analytical Problem Solving:** Research-oriented approach to complex domains, including Post-Quantum Cryptography and Vedic Mathematics.
- **Strategic Communication:** Ability to translate complex technical concepts for non-technical clients.

Leadership / Extracurricular

Indian Knowledge from Vedic Insights (IKVI) 2025

Shobhit University, 2025

- Authored and presented a research paper titled "**Integrating Vedic Mathematics in Post-Quantum Cryptography**" at the IKVI 2025. The abstract has been published, and the full research paper is currently under review.

Winner, Internal Smart India Hackathon 2025

Shobhit University, 2025

- Implemented on-device image processing to automatically blur inappropriate visuals, ensuring user privacy while preventing early exposure to harmful material.

Peer-to-Peer Workshop on MERN Development

Shobhit University, 2025

- Conducted hands-on web development sessions — fostering collaborative learning, practical implementation, and technical confidence.

Profiles: leetcode.com/u/kshwebsites | hackerrank.com/profile/kshwebsites